

# Out-of-Distribution Detection in Computer Vision

Comparative benchmark with a simplified STOOD-X statistical approach

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# Problem and Objective

**Problem.** A classifier trained on CIFAR-10 can still assign high confidence to images outside its training distribution.

**Objective.** Detect whether an input image is In-Distribution (ID) or Out-of-Distribution (OOD), while keeping the trained classifier fixed.

## Project focus.

- Train ResNet-18 on CIFAR-10.
- Benchmark post-hoc OOD detectors.
- Partially replicate the feature-space intuition of STOOD-X.



CIFAR-10 is the known distribution.

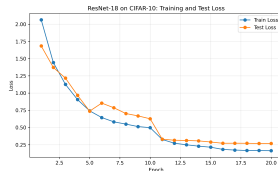
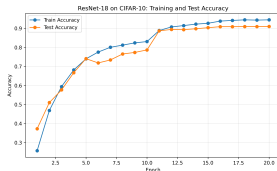
# Experimental Setup

## Datasets

- ID: CIFAR-10.
- OOD: CIFAR-100, SVHN, MNIST.
- All inputs normalized with CIFAR-10 statistics.

## Classifier

- ResNet-18 adapted for  $32 \times 32$  images.
- 20 training epochs.
- Best CIFAR-10 test accuracy: **91.03%**.



## Metrics

- AUROC, AUPR-IN, AUPR-OUT: higher is better.
- FPR95: lower is better.

# Methods Compared

## Logit/confidence based

- **MSP**: maximum softmax probability.
- **Energy**: log-sum-exp score over logits.
- **ODIN**: temperature scaling plus input perturbation.

These methods use the classifier output directly.

## Feature-distance based

- **Mahalanobis**: distance to class feature means.
- **KNN**: distance to the ID feature bank,  $k = 50$ .
- **STOOD-X simplified**: empirical p-value from KNN distances.

These methods use penultimate-layer representations.

**Common convention:** high score = more likely ID; low score = more likely OOD.

# Simplified STOOD-X Procedure

- 1 Extract ResNet-18 features from CIFAR-10 training images.
- 2 Build an ID feature bank.
- 3 For each ID training sample, compute the distance to its  $k$ -th nearest neighbor.
- 4 Use these distances as the reference ID distribution.
- 5 For a new sample, convert its KNN distance into an empirical p-value.

This is a partial replication: the full STOOD-X explainability stage and the original Wilcoxon-Mann-Whitney test were not implemented.

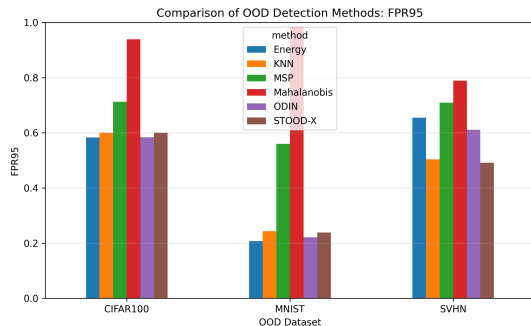
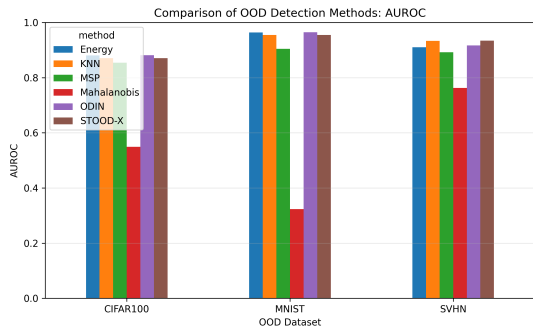
$$p(x) = \frac{1}{N} \sum_{i=1}^N \mathbb{I}(d_i^{ref} \geq d(x))$$

## Interpretation

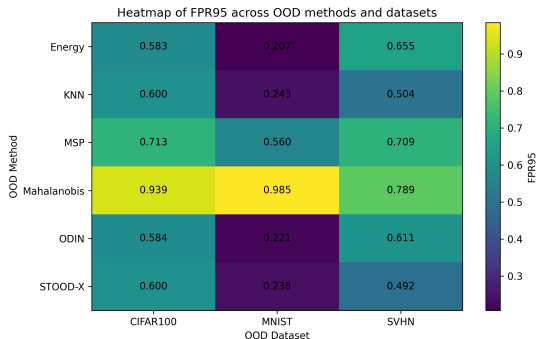
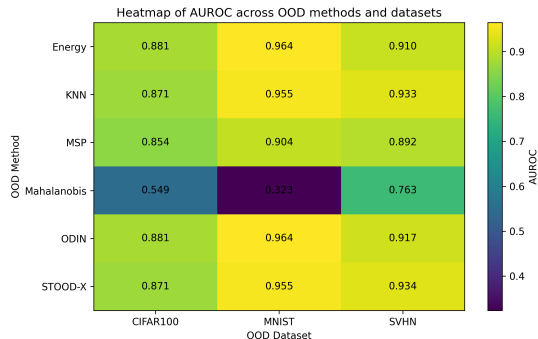
- High p-value: close to ID feature geometry.
- Low p-value: statistically unusual feature distance.

# Quantitative Benchmark

| OOD Dataset | Best AUROC             | Best FPR95             | Strongest Pattern     |
|-------------|------------------------|------------------------|-----------------------|
| CIFAR-100   | ODIN: <b>0.8810</b>    | Energy: <b>0.5825</b>  | hardest near-OOD case |
| MNIST       | ODIN: <b>0.9643</b>    | Energy: <b>0.2073</b>  | easiest far-OOD case  |
| SVHN        | STOOD-X: <b>0.9339</b> | STOOD-X: <b>0.4916</b> | feature distances win |

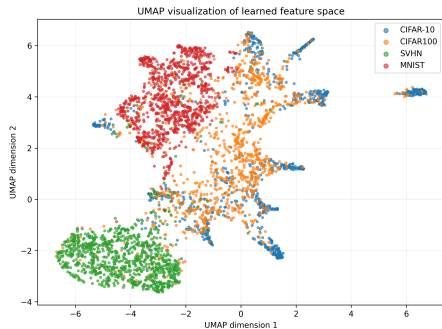
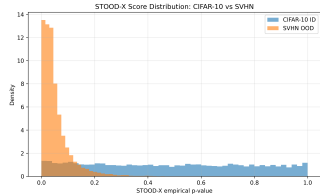
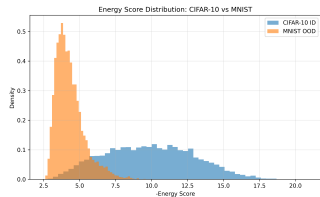


# Full Result Matrix



- ODIN and Energy are strongest on CIFAR-100 and MNIST.
- KNN and simplified STOOD-X are very close, because both rely on feature-space distances.
- Mahalanobis underperforms, especially on MNIST and CIFAR-100.

# Score and Feature-Space Evidence



- Good detectors separate ID and OOD score distributions.
- UMAP confirms that feature geometry contains useful OOD information.



# Main Conclusions

- The ResNet-18 classifier provides a strong feature space for post-hoc OOD detection.
- ODIN and Energy are the best overall logit-based methods.
- Simplified STOOD-X is strongest on SVHN and competitive on MNIST.
- Feature-distance methods are useful when OOD samples are not simply low-confidence inputs.
- The project produces reproducible outputs: metrics tables, ROC curves, score distributions, heatmaps, t-SNE/UMAP and a final report.

**Limitations.** Single architecture, single training run, three OOD datasets, and only a partial STOOD-X replication.

**Takeaway:** OOD detection improves when classifier confidence is combined with feature-space evidence.